

John Abednego Jilima (22424153)
Supervisor: Prof. Solomon Mensah
Department of Computer Science, University of Ghana

Introduction

Defect prediction needs history. New teams and greenfield products have none, so they must borrow a model from another project.

Accuracy then depends heavily on which project supplies the training data. Zimmermann et al. (2009) found transfer succeeded in only 4% of 622 project pairs.

The bellwether hypothesis (Krishna & Menzies, 2019) holds that each community contains one exemplar that transfers well to all members. A widely assumed explanation is that such a project sits near its community centroid, a premise that underpins training-data selection (Herbold, 2013) and hierarchical identification (Majumder et al., 2021).

That premise has not previously been tested against measured transfer outcomes.

Aim

To establish by direct measurement which standing assumptions of bellwether-based cross-project prediction survive scrutiny, and to supply corrected mechanisms where they do not.

Methods

Data. 30 publicly archived projects across three collections with disjoint metric suites: PROMISE Jureczko (20 releases, 20 metrics), AEEEM (5 projects, 61 metrics), NASA MDP in the Shepperd et al. (2013) cleaned variant (5 projects, 21 metrics). Every source file fixed by SHA-256 digest.

Framework. Five modules: curation, feature harmonisation, bellwether identification, a weighted random-forest and logistic-regression ensemble, and stability monitoring.

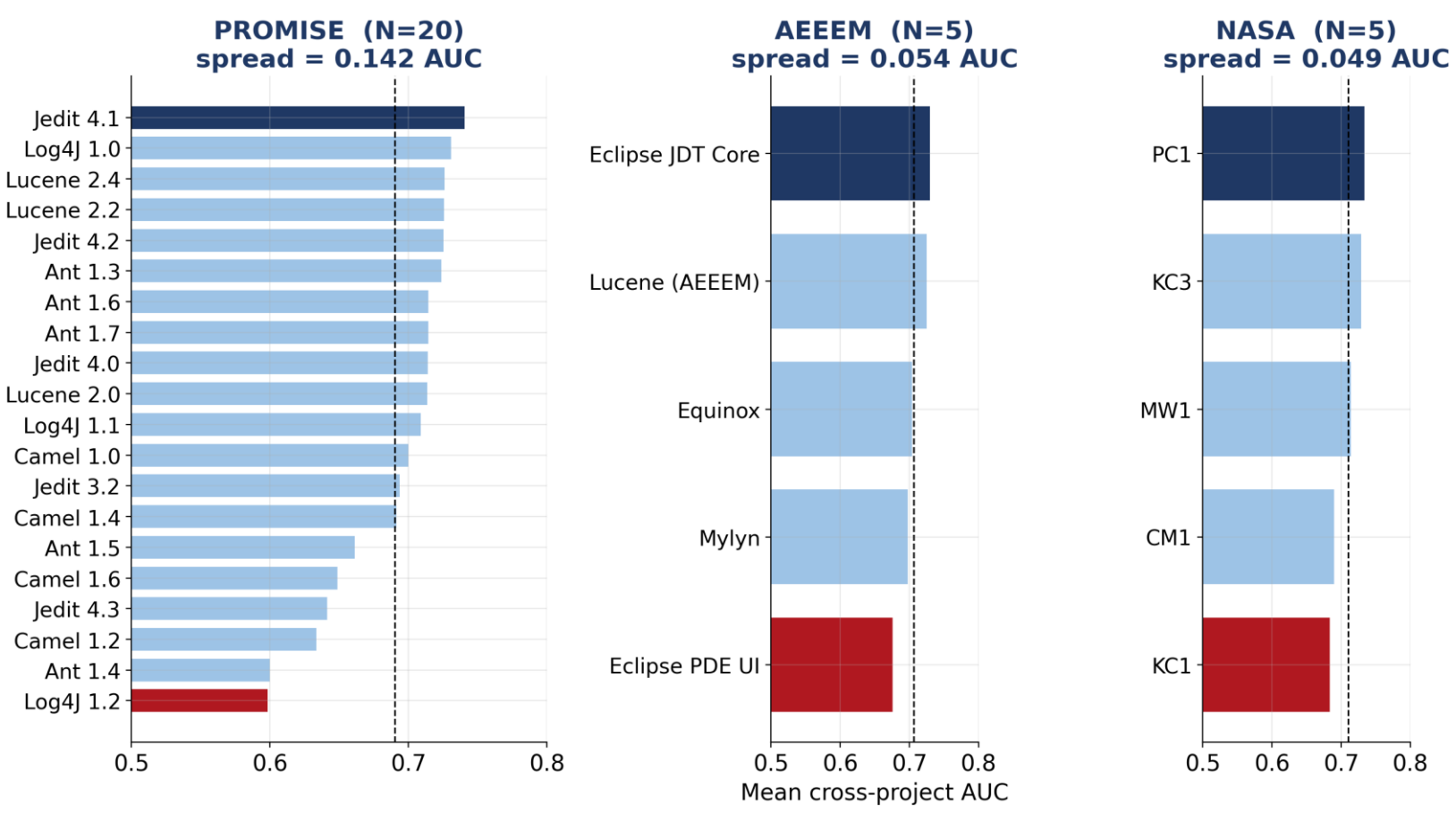
Identification. Three strategies compared: exhaustive round-robin, a clustering variant following the design principle of Majumder et al. (2021), and a probe variant introduced here that scores each candidate against $\lceil \log_2 N \rceil$ random targets before a restricted round-robin.

Baselines. Within-project prediction, global pooling, TCA+, Transfer Naive Bayes and Heterogeneous Defect Prediction, each in random-source and bellwether-source configurations.

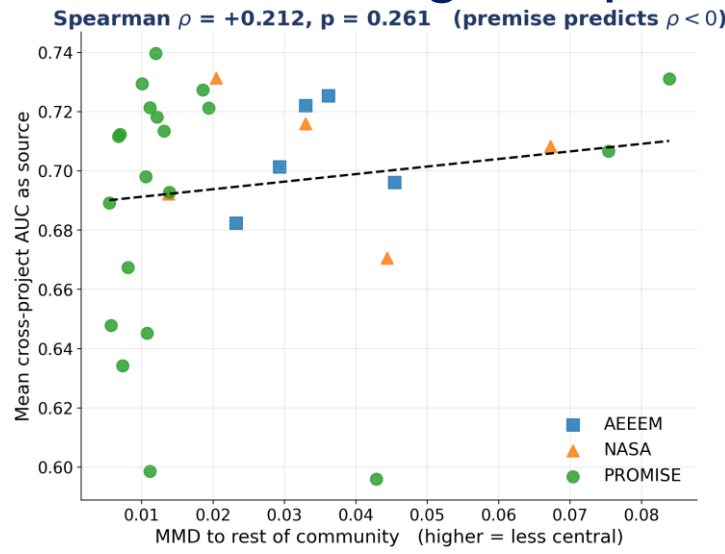
Protocol. Leave-one-project-out, so no method sees target labels while selecting a source. Over 3,800 model fits. Wilcoxon signed-rank with Benjamini-Hochberg correction, Cliff's delta and Scott-Knott ESD.

Results

Finding 1. The bellwether effect is real. Source choice alone moves mean cross-project AUC by 0.144 in PROMISE, 0.054 in AEEEM and 0.049 in NASA. The effect replicates across all three metric suites.



Finding 3. The distributional-centrality premise is not supported. Across 30 projects the Spearman correlation between maximum mean discrepancy to the community and mean cross-project AUC across thirty seeds is +0.212 at $p = 0.261$, opposite in sign to the prediction. No identified bellwether is the most central member of its collection (ranks 12/20, 4/5, 2/5). No correlation reaches significance, so this is absence of support rather than evidence for a contrary mechanism. With five projects in two of three collections the individual tests are severely underpowered, and no weight is placed on the direction.



Collection	Best	Worst	Spread
PROMISE (N=20)	0.741	0.598	0.144
AEEEM (N=5)	0.730	0.676	0.054
NASA MDP (N=5)	0.733	0.684	0.049

Finding 2. Replication of a published result.

The AEEEM curation reproduces the defective-instance counts published by Krishna and Menzies exactly. Following their thirty-seed protocol the full experiment was repeated across thirty seeds, giving eighty-nine seed-collection runs. LC is selected as the AEEEM bellwether in 20 of 30 runs, which is the identity they report; a single seed returns JDT, a minority outcome separated by 0.0033 AUC. PROMISE selects Jedit 4.1 in 29 of 29 runs and NASA selects PC1 in 29 of 30. Where candidates sit thousandths of an AUC apart, seed repetition is a precondition for any claim about identity.

Finding 4. Comparisons must be made on a common target set.

Across 89 runs the clustering variant recovers the exhaustive bellwether in 22.2% of cases against a pooled chance rate of 18.6%, and on the twenty-project collection it falls below chance at 1.4% against 5.3%, forfeiting 0.0214 AUC (adjusted $p = 1.2e-14$, large effect). The cause is a defect in this study's own implementation: cluster representatives were scored on target sets that differed between them and were never rescored on a common one. Repairing the comparison alone recovers 93.6%. The probe variant does rescore its finalists, raising agreement to 75.9% and returning 98.7% of exhaustive accuracy for 44% of its cost: a trade-off rather than parity.

Strategy	Agreement	AUC	vs exhaustive	Cost
Exhaustive	100%	0.7206	reference	10,638
Clustering	22.2%	0.6991	$p = 3.9e-15$	5,264
Probe (this work)	75.9%	0.7162	$p = 5.1e-4$	5,341

References

- Krishna, R., & Menzies, T. (2019). Bellwethers: A baseline method for transfer learning. IEEE TSE, 45(11), 1081-1105.
- Majumder, S., Krishna, R., Xia, T., & Menzies, T. (2021). How to GENERALize across many software projects. arXiv:1911.04250.
- Shepperd, M., Song, Q., Sun, Z., & Mair, C. (2013). Data quality: Some comments on the NASA software defect datasets. IEEE TSE, 39(9), 1208-1215.
- Nam, J., Fu, W., Kim, S., Menzies, T., & Tan, L. (2018). Heterogeneous defect prediction. IEEE TSE, 44(9), 874-896.
- Zimmermann, T., Nagappan, N., Gall, H., Giger, E., & Murphy, B. (2009). Cross-project defect prediction. ESEC/FSE, 91-100.
- Herbold, S. (2013). Training data selection for cross-project defect prediction. PROMISE 2013.

Observations

The framework significantly outperforms every transfer baseline lacking a source-selection oracle (Cliff's delta up to 0.398, corrected $p < 0.001$), and matches exhaustive bellwether search at 45% of its evaluation cost. Within-project prediction remains significantly superior (0.783 AUC, $p = 0.0004$), and global pooling ties on AUC while reaching only 0.275 recall against 0.432 for bellwether transfer. Bellwether source quality generalises across tasks ($\rho = +0.633$, $p = 0.003$). Identification results are repeated across thirty seeds. Under a prospective protocol identity is retained in only 2 of 8 comparisons while performance does not decay ($p = 0.742$), locating temporal validity in re-identification rather than tenure.

Conclusion

The bellwether effect is real and reproduces across three collections, and the AEEEM result replicates a published identification. The assumption used to explain the effect does not survive measurement, and an engine resting on it fails with a large effect. Sparse direct measurement recovers most of that shortfall at 44% of the cost.

Data and code: all 30 datasets are publicly archived and fixed by SHA-256 digest; every figure and table is generated from stored result files.